

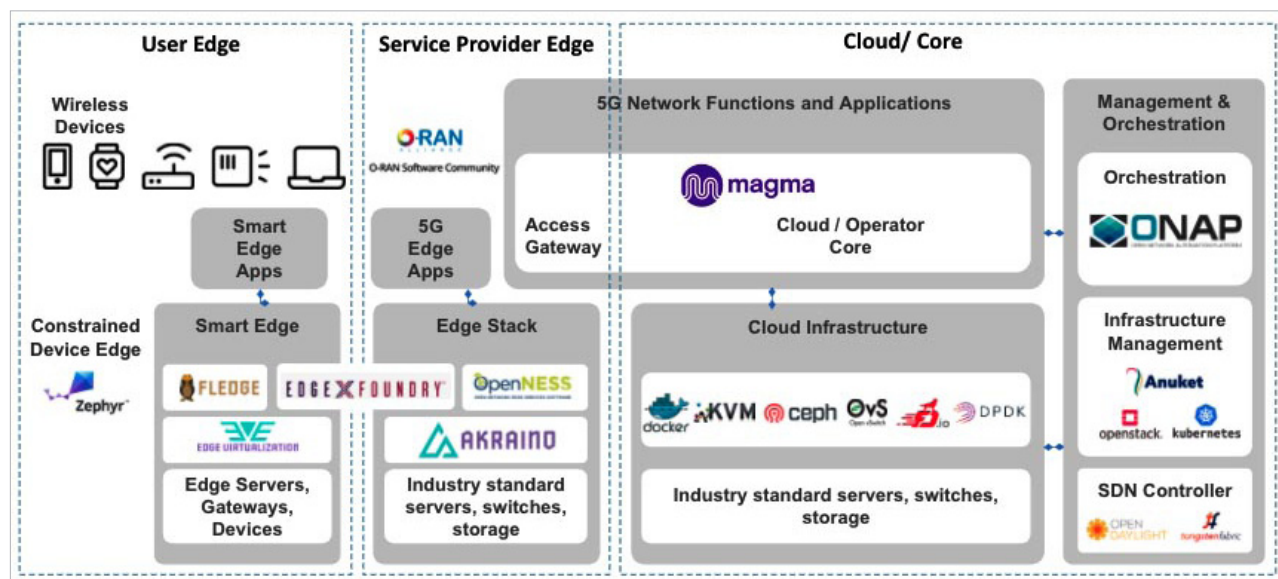


Figure 1: LF open source component projects for 5G

End-to-end open source software collaboration

LF Networking has a broad set of projects that make up the diverse and evolving open source networking stack. It provides platforms and building blocks across the networking industry that enable rapid interoperability, deployment and adoption, and is the nexus for 5G innovation and integration. It has now tapped into industry efforts to structure a new initiative to develop 5G Super Blueprints for the ecosystem. Major integrations between the building blocks are now underway—between ONAP and ORAN, Akraino and Magma, Anuket and Kubernetes, and more.

The 5G Super Blueprint is what is going to integrate a set of projects, all the way from access to core.

‘Super’ means that we are integrating multiple projects and umbrellas (such as LF Edge, Magma, CNCF, O-RAN Alliance, LF Energy, and more) with an end-to-end structure for the underlying facilities and application layers. This end-to-end combination allows leading market usage cases, such as repaired wireless, mobile broadband, personal 5G, multi-

access, IoT, voice services, network slicing, and more. In short, 5G Super Blueprints help to produce end-to-end 5G options.

“With the availability of customised solutions, there is a tremendous opportunity for the market. This is how the verticals are taking advantage of edge computing. Now healthcare and smart cities are towards the lower end of the spectrum. But it will need a few regulations and process controls before adoption happens,” says Joshipura.

LF edge projects

LF edge project is standard vocabulary for edge users. There are two types of edges — user edge and service provider edge.

A user edge is dedicated and operated by the user. A service provider edge is shared as a service, but not all usages are the same. You can either have a very constrained device edge, which could be microcontroller based, or you can have a gateway type edge.

And what separates the two is the last mile network. “Edge compute is for applications that need 20 milliseconds or less latency,” he says.

Joshipura says the edge project stack is a very important attribute. He explains with an example: “If a sensor wakes up and dumps data to an Amazon cloud every week, it’s an IoT application. It’s not an edge application and vice versa. Hence, the LF edge as an umbrella is unifying the frameworks.”

Akraino R4 blueprints

Akraino is a blueprint concept. The open community comes together to define a use case. Then through open governance, it tests the use case and interoperability — all the way from hardware up to the application. It then submits the results so that developers can replicate it.

“That’s a blueprint, a declarative configuration, with CI/CD documentation. The one that has become very popular with telcos is PCEI (public cloud edge interface),” says Joshipura.

The question is: how do the public cloud and the service provider at the edge have a common set of interfaces. Joshipura says there are no standards that exist.